

SECTION A

Answer all questions in the spaces provided.

1. When aqueous iodide ions are added to an aqueous solution of lead(II) nitrate, a precipitate is formed.

(a) Give the colour of the precipitate. [1]

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(b) Write an **ionic** equation for the reaction that occurs, including state symbols. [1]

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2. Give an example of a transition metal used as a catalyst. You should name the metal and identify the reaction it catalyses. [1]

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3. Calcium chloride, CaCl_2 , is soluble in water. Use the values below to explain why this is the case. [2]



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(c) Nitrogen dioxide is used in the production of nitric acid. It is produced in two stages.



- (i) Explain why the use of a catalyst is essential in industrial processes involving exothermic equilibria such as stage 1. [2]

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- (ii) A student attempted to perform stage 2 using a temperature of 450 K and a pressure of 5 atm. He obtained only a small yield. Suggest how the yield could be improved, explaining your reasons. [4]

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